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Section-6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

9. Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut  skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create the Directory Navigation Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano directory_navigation.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 directory_navigation.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define INPUT_SIZE 200

int main()
{
    char input[INPUT_SIZE];
    char previous_dir[PATH_MAX];
    char current_dir[PATH_MAX];

    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        return 1;
    }

    strcpy(previous_dir, current_dir);

    while (1)
    {
        printf("nav> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
        {
            break;
        }

        if (strcmp(input, "pwd") == 0)
        {
            if (getcwd(current_dir, sizeof(current_dir)) != NULL)
            {
                printf("Current directory: %s\n", current_dir);
            }
        }
    }
}
```

Help	Write Out	Where Is	Cut	Execute	Location	Undo
Exit	Read File	Replace	Paste	Justify	Go To Line	Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 directory_navigation.c *
}
else
{
    perror("getcwd");
}

continue;
}

if (strncmp(input, "cd ", 3) == 0)
{
    char *path = input + 3;

    if (strlen(path) == 0)
    {
        printf("Error: Path is empty.\n");
        continue;
    }

    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        continue;
    }

    strcpy(previous_dir, current_dir);

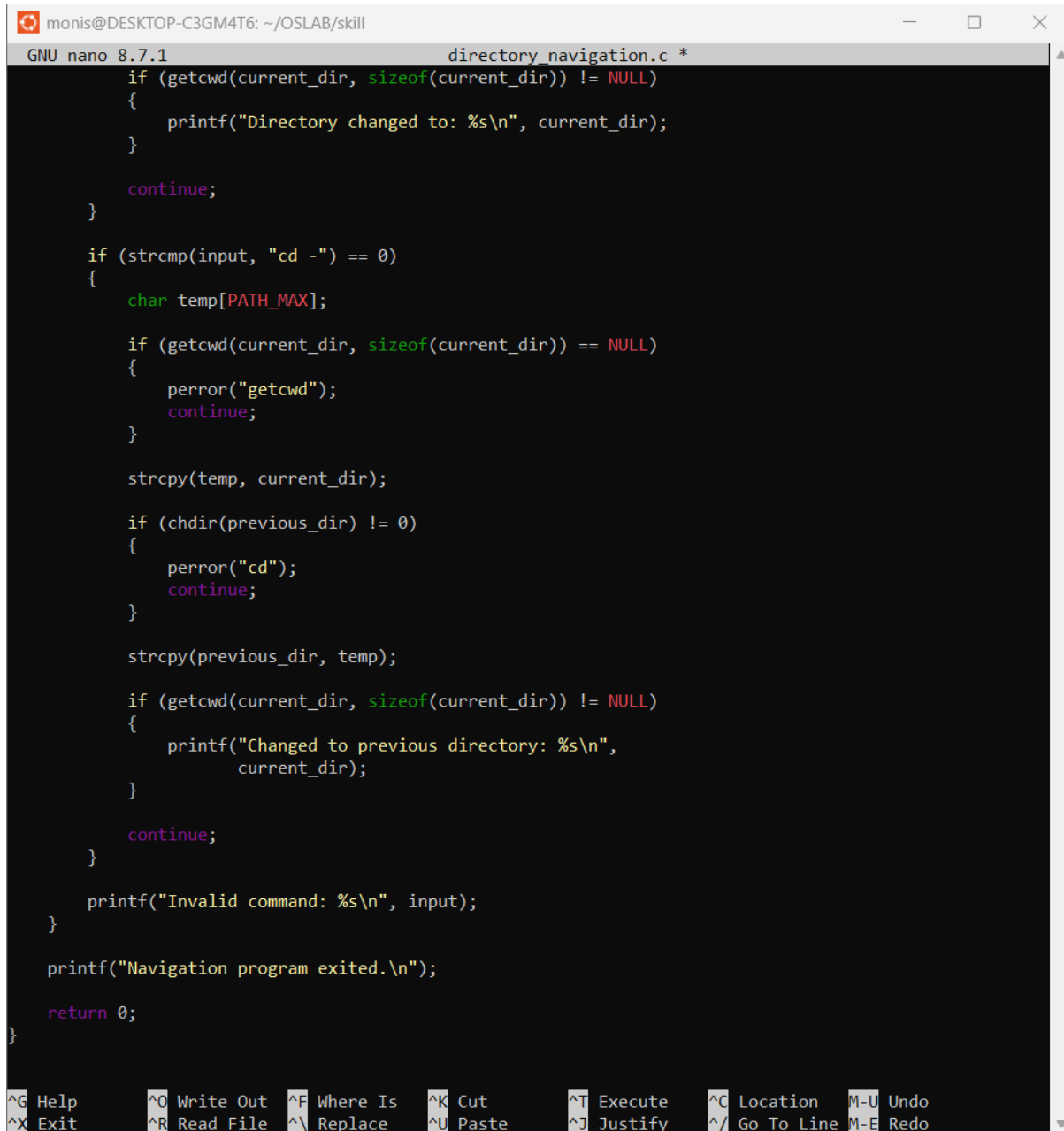
    if (chdir(path) != 0)
    {
        perror("cd");
        continue;
    }

    if (getcwd(current_dir, sizeof(current_dir)) != NULL)
    {
        printf("Directory changed to: %s\n", current_dir);
    }

    continue;
}

if (strcmp(input, "cd -") == 0)
{
    char temp[PATH_MAX];

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 directory_navigation.c *
    if (getcwd(current_dir, sizeof(current_dir)) != NULL)
    {
        printf("Directory changed to: %s\n", current_dir);
    }

    continue;
}

if (strcmp(input, "cd -") == 0)
{
    char temp[PATH_MAX];

    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        continue;
    }

    strcpy(temp, current_dir);

    if (chdir(previous_dir) != 0)
    {
        perror("cd");
        continue;
    }

    strcpy(previous_dir, temp);

    if (getcwd(current_dir, sizeof(current_dir)) != NULL)
    {
        printf("Changed to previous directory: %s\n",
            current_dir);
    }

    continue;
}

printf("Invalid command: %s\n", input);
}

printf("Navigation program exited.\n");

return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

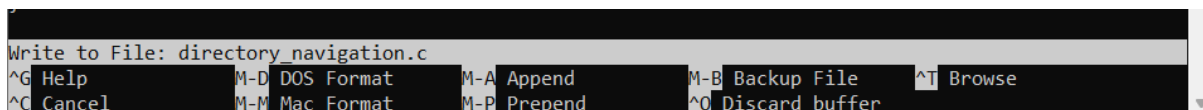
Save the file

Press:

Ctrl + O

Enter

Ctrl + X



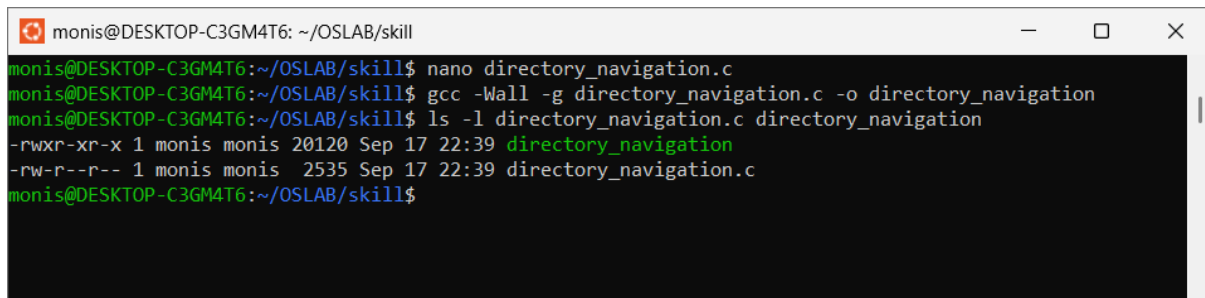
```
Write to File: directory_navigation.c
^G Help      M-D DOS Format M-A Append   M-B Backup File ^T Browse
^C Cancel     M-M Mac Format M-P Prepend  ^Q Discard buffer
```

Step 3 — Compile the Directory Navigation Program

Now run:

```
gcc -Wall -g directory_navigation.c -o directory_navigation
```

If there are no errors, check that the source file and executable were created:
`ls -l directory_navigation.c directory_navigation`



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano directory_navigation.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g directory_navigation.c -o directory_navigation
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l directory_navigation.c directory_navigation
-rwxr-xr-x 1 monis monis 20120 Sep 17 22:39 directory_navigation
-rw-r--r-- 1 monis monis  2535 Sep 17 22:39 directory_navigation.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 — Test Changing Directories

Now run:

`./directory_navigation`

You should see:

`nav>`

At the `nav>` prompt, type:

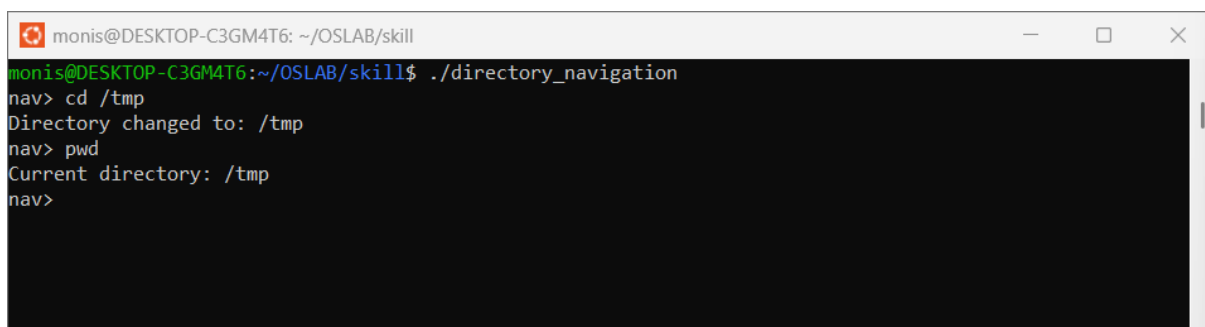
`cd /tmp`

Press **Enter**.

Then type:

`pwd`

Press **Enter**.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_navigation
nav> cd /tmp
Directory changed to: /tmp
nav> pwd
Current directory: /tmp
nav>
```

Step 5 — Test Previous Directory Navigation

The program should still be running at:

`nav>`

You are currently in `/tmp`.

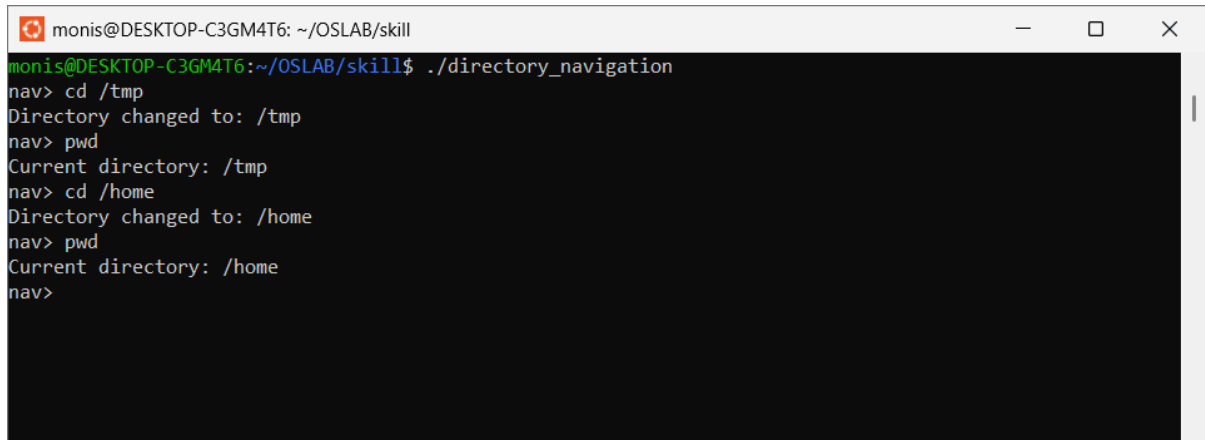
Now type:

`cd /home`

Press Enter.

Then type:

`pwd`

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user has entered './directory_navigation'. The program output shows a navigation menu 'nav>' with two options: 'cd /tmp' and 'cd /home'. The user selected 'cd /tmp', and the program responded with 'Directory changed to: /tmp' and 'Current directory: /tmp'. Then the user selected 'cd /home', and the program responded with 'Directory changed to: /home' and 'Current directory: /home'. The prompt is now 'nav>' again.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./directory_navigation
nav> cd /tmp
Directory changed to: /tmp
nav> pwd
Current directory: /tmp
nav> cd /home
Directory changed to: /home
nav> pwd
Current directory: /home
nav>
```

Step 6 — Test Invalid Directory Path

The program should still be running at:

`nav>`

Now type:

`cd /directory_that_does_not_exist`

Press Enter.

Now verify that the program is still running by typing:

`pwd`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_navigation
nav> cd /tmp
Directory changed to: /tmp
nav> pwd
Current directory: /tmp
nav> cd /home
Directory changed to: /home
nav> pwd
Current directory: /home
nav> cd -
cd: No such file or directory
nav>
Invalid command:
nav> cd -
cd: No such file or directory
nav> pwd
Current directory: /home
nav> cd -
cd: No such file or directory
nav> cd /directory_that_does_not_exist
cd: No such file or directory
nav> pwd
Current directory: /home
nav>
```

Step 7 — Finish Part 1 and Start Built-in Dispatch

First, finish the directory-navigation program.

At:

nav>

type:

exit

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_navigation
nav> cd /tmp
Directory changed to: /tmp
nav> pwd
Current directory: /tmp
nav> cd /home
Directory changed to: /home
nav> pwd
Current directory: /home
nav> cd -
cd: No such file or directory
nav>
Invalid command:
nav> cd -
cd: No such file or directory
nav> pwd
Current directory: /home
nav> cd -
cd: No such file or directory
nav> cd /directory_that_does_not_exist
cd: No such file or directory
nav> pwd
Current directory: /home
nav> exit
Navigation program exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 8 — Create the Built-in Dispatch Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano builtin_dispatch.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 builtin_dispatch.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define INPUT_SIZE 200
#define MAX_ARGS 20

typedef void (*builtin_func)(char **);

void builtin_pwd(char **args)
{
    char cwd[1024];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
    else
        perror("pwd");
}

void builtin_cd(char **args)
{
    if (args[1] == NULL)
    {
        printf("cd: missing argument\n");
        return;
    }

    if (chdir(args[1]) != 0)
        perror("cd");
}

void builtin_help(char **args)
{
    printf("Built-in commands:\n");
    printf("  pwd  - show current directory\n");
    printf("  cd   - change directory\n");
    printf("  help - show available commands\n");
    printf("  exit - exit the shell\n");
}

struct Builtin
{
    char *name;
    builtin_func function;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 builtin_dispatch.c *
};
builtin_func function;

struct Builtin builtins[] =
{
    {"pwd", builtin_pwd},
    {"cd", builtin_cd},
    {"help", builtin_help}
};

int builtin_count = sizeof(builtins) / sizeof(builtins[0]);

int main()
{
    char input[INPUT_SIZE];
    char *args[MAX_ARGS];
    int running = 1;

    printf("Built-in Dispatch Shell\n");

    while (running)
    {
        printf("dispatch> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

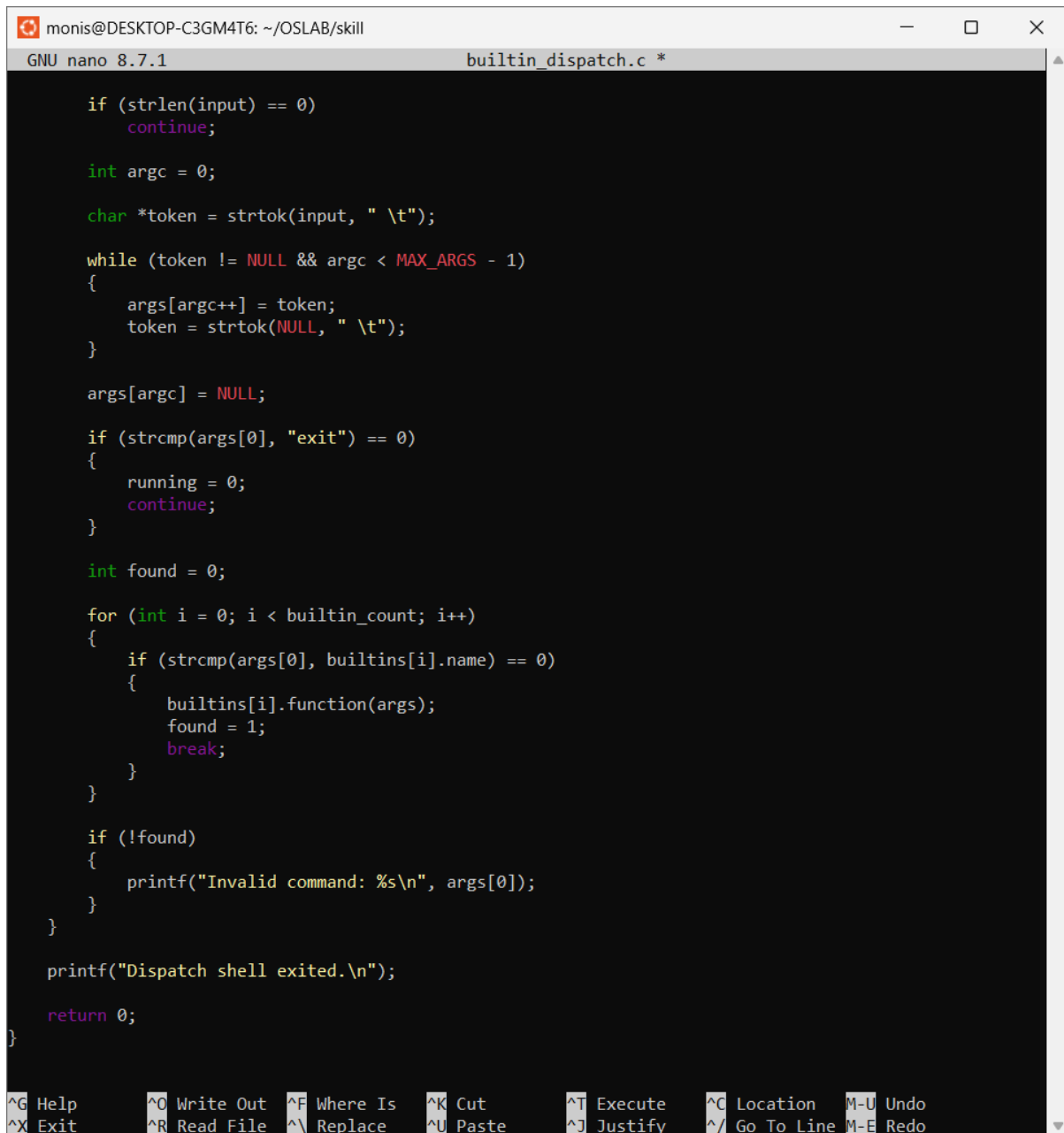
        int argc = 0;

        char *token = strtok(input, " \t");

        while (token != NULL && argc < MAX_ARGS - 1)
        {
            args[argc++] = token;
            token = strtok(NULL, " \t");
        }

        args[argc] = NULL;
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^/_ Go To Line	M-E Redo



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 builtin_dispatch.c *

    if (strlen(input) == 0)
        continue;

    int argc = 0;

    char *token = strtok(input, " \t");

    while (token != NULL && argc < MAX_ARGS - 1)
    {
        args[argc++] = token;
        token = strtok(NULL, " \t");
    }

    args[argc] = NULL;

    if (strcmp(args[0], "exit") == 0)
    {
        running = 0;
        continue;
    }

    int found = 0;

    for (int i = 0; i < builtin_count; i++)
    {
        if (strcmp(args[0], builtins[i].name) == 0)
        {
            builtins[i].function(args);
            found = 1;
            break;
        }
    }

    if (!found)
    {
        printf("Invalid command: %s\n", args[0]);
    }

    printf("Dispatch shell exited.\n");

    return 0;
}
```

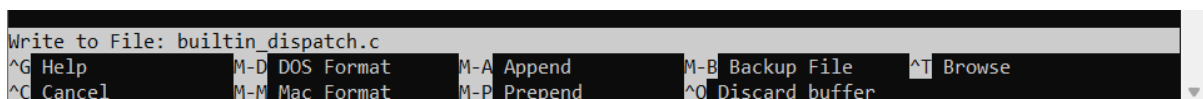
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo

Save:

Ctrl + O

Enter

Ctrl + X



Write to File: builtin_dispatch.c

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^Q Discard buffer

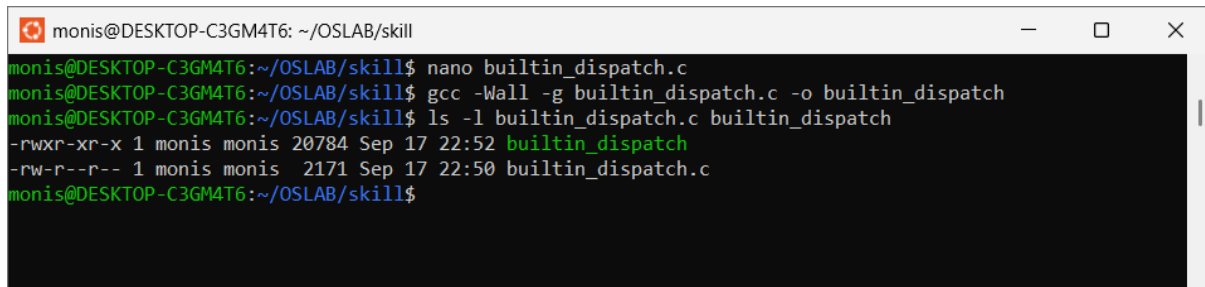
Step 9 — Compile the Built-in Dispatch Program

Run:

```
gcc -Wall -g builtin_dispatch.c -o builtin_dispatch
```

If there are **no errors or warnings**, check the files:

```
ls -l builtin_dispatch.c builtin_dispatch
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The user has executed the following commands: 'nano builtin_dispatch.c', 'gcc -Wall -g builtin_dispatch.c -o builtin_dispatch', and 'ls -l builtin_dispatch.c builtin_dispatch'. The output of the last command shows two files: 'builtin_dispatch' with permissions '-rwxr-xr-x' and 'builtin_dispatch.c' with permissions '-rw-r--r--'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ nano builtin_dispatch.c
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ gcc -Wall -g builtin_dispatch.c -o builtin_dispatch
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ls -l builtin_dispatch.c builtin_dispatch
-rwxr-xr-x 1 monis monis 20784 Sep 17 22:52 builtin_dispatch
-rw-r--r-- 1 monis monis 2171 Sep 17 22:50 builtin_dispatch.c
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$
```

Step 10 — Run the Built-in Dispatch Program

Run:

```
./builtin_dispatch
```

You should see:

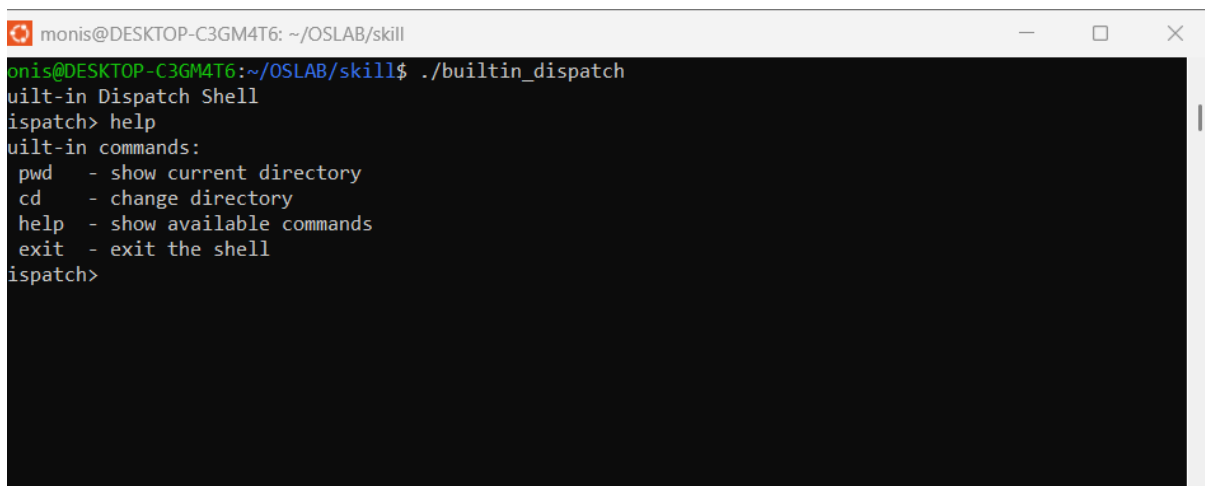
Built-in Dispatch Shell

dispatch>

Now type:

help

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The user has executed './builtin_dispatch', which has started the 'Built-in Dispatch Shell'. The prompt is 'dispatch>'. The user has entered 'help', and the shell has displayed a list of built-in commands: 'pwd' (show current directory), 'cd' (change directory), 'help' (show available commands), and 'exit' (exit the shell).

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
pwd  - show current directory
cd   - change directory
help - show available commands
exit - exit the shell
dispatch>
```

Step 11 — Test the pwd Built-in

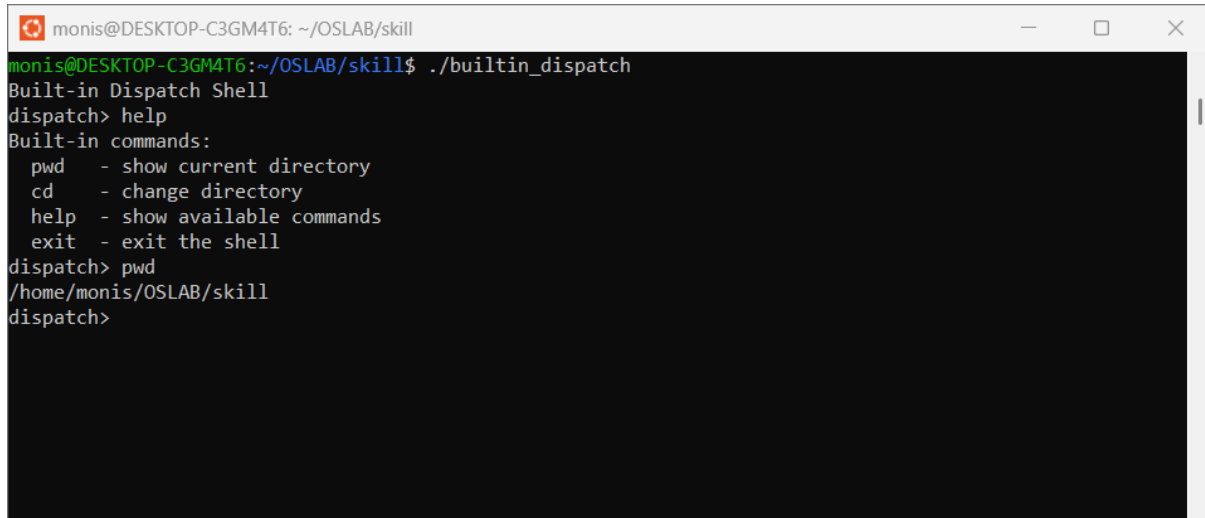
The program should still be running at:

dispatch>

Type:

pwd

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './builtin_dispatch', and the prompt changes to 'Built-in Dispatch Shell' and 'dispatch>'. The user enters 'help', and the prompt changes to 'Built-in commands:'. The user enters 'pwd', and the prompt changes to 'dispatch>'. The output is '/home/monis/OSLAB/skill'. The user enters 'dispatch>' again.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd - show current directory
  cd  - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch>
```

Step 12 — Test the cd Built-in and State Maintenance

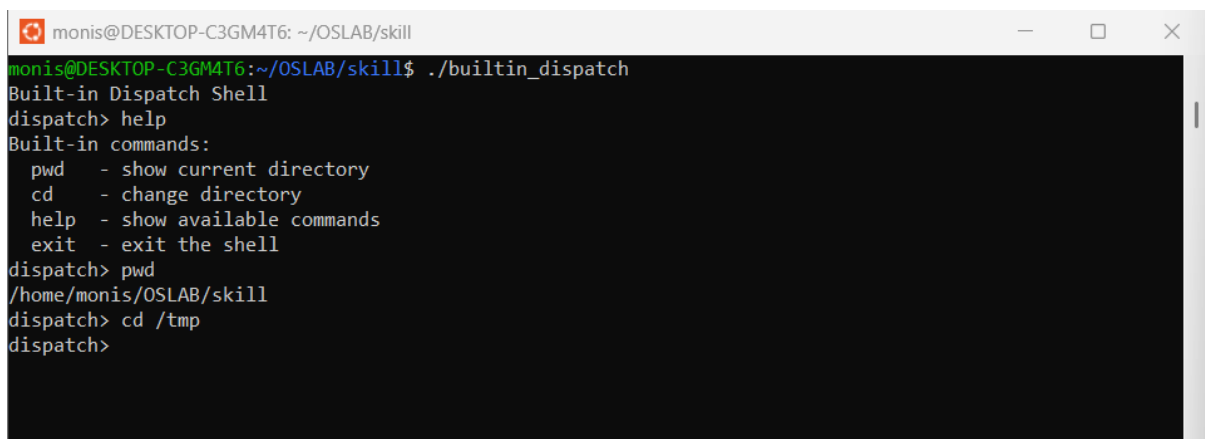
The program should still be running at:

dispatch>

First type:

cd /tmp

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$'. The user enters './builtin_dispatch', and the prompt changes to 'Built-in Dispatch Shell' and 'dispatch>'. The user enters 'help', and the prompt changes to 'Built-in commands:'. The user enters 'pwd', and the prompt changes to 'dispatch>'. The output is '/home/monis/OSLAB/skill'. The user enters 'dispatch> cd /tmp', and the prompt changes to 'dispatch>'. The user enters 'dispatch>' again.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd - show current directory
  cd  - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch> cd /tmp
dispatch>
```

Step 13 — Test an Invalid Command

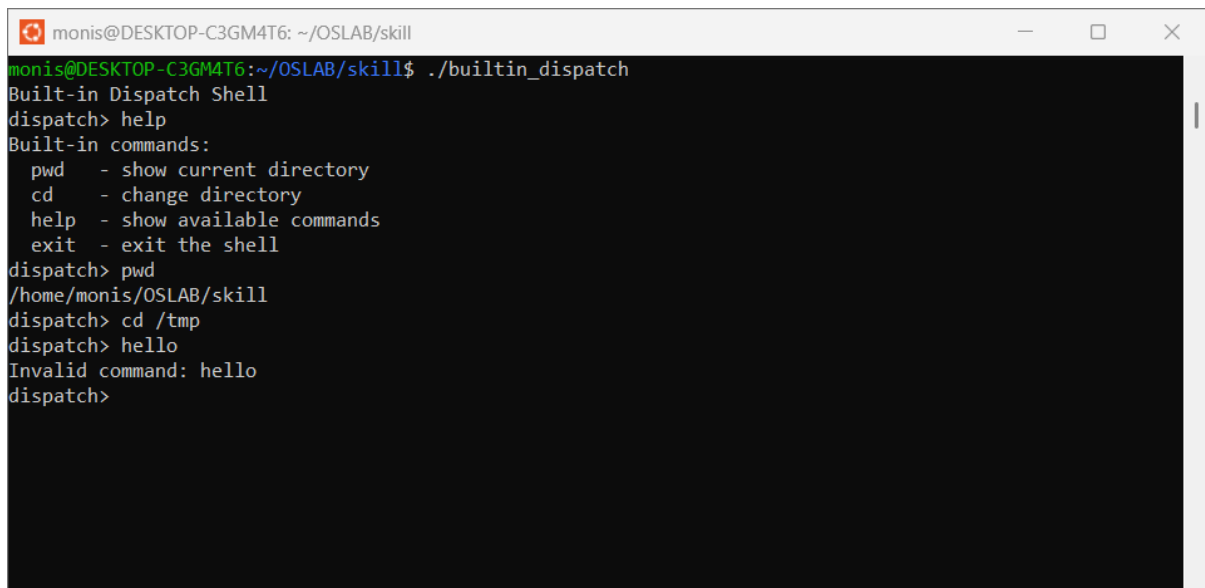
At the same:

dispatch>

type:

hello

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$' and the command entered is './builtin_dispatch'. The output shows the 'Built-in Dispatch Shell' and a list of built-in commands: 'pwd' (show current directory), 'cd' (change directory), 'help' (show available commands), and 'exit' (exit the shell). The user then enters 'dispatch> pwd' which outputs '/home/monis/OSLAB/skill', 'dispatch> cd /tmp', and 'dispatch> hello' which results in 'Invalid command: hello'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd  - show current directory
  cd   - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch> cd /tmp
dispatch> hello
Invalid command: hello
dispatch>
```

Step 14 — Test cd Error Handling

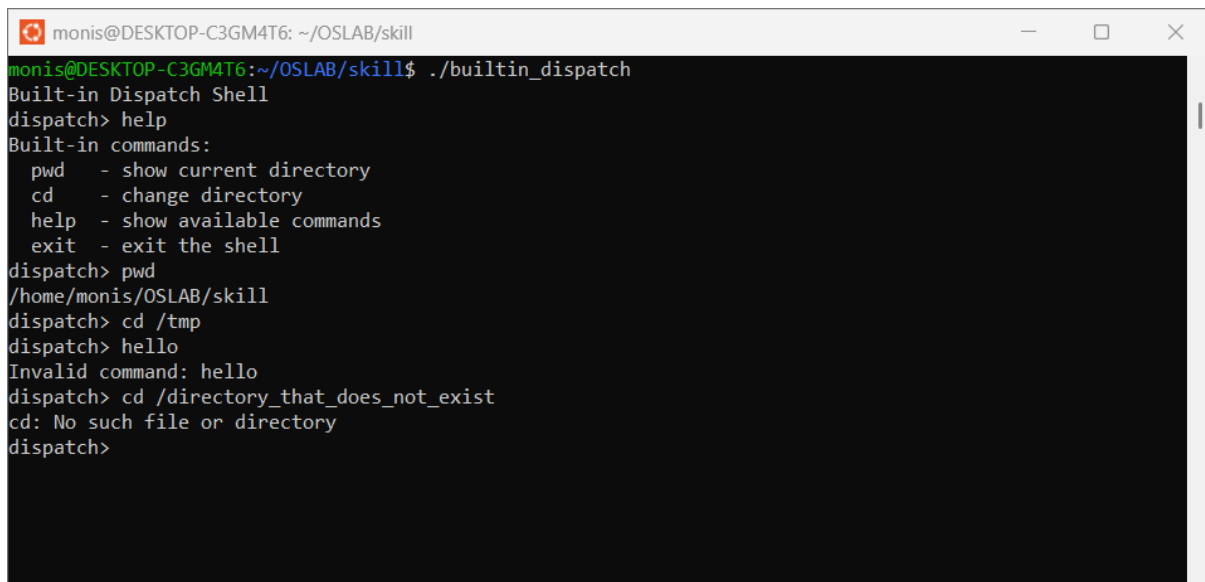
At:

dispatch>

type:

cd /directory_that_does_not_exist

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$' and the command entered is './builtin_dispatch'. The output shows the 'Built-in Dispatch Shell' and a list of built-in commands. The user then enters 'dispatch> pwd' which outputs '/home/monis/OSLAB/skill', 'dispatch> cd /tmp', 'dispatch> hello' which results in 'Invalid command: hello', and 'dispatch> cd /directory_that_does_not_exist' which results in 'cd: No such file or directory'.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd  - show current directory
  cd   - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch> cd /tmp
dispatch> hello
Invalid command: hello
dispatch> cd /directory_that_does_not_exist
cd: No such file or directory
dispatch>
```

Step 15 — Test cd with No Argument

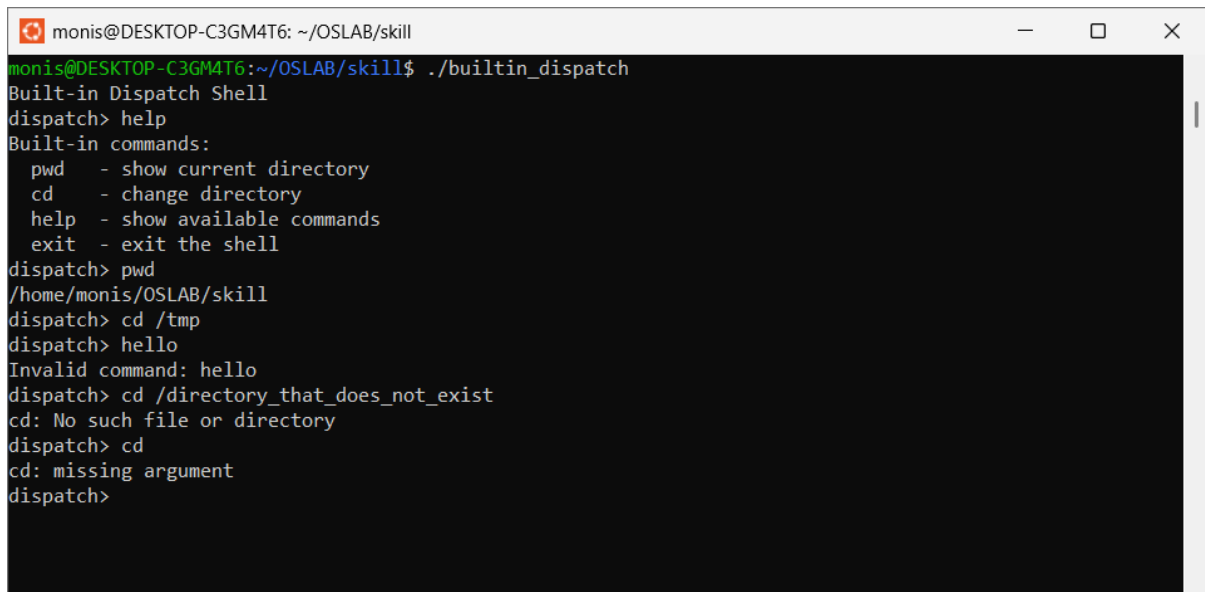
At:

dispatch>

type:

cd

Press **Enter**.

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill'. The prompt is 'monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$' and the command entered is './builtin_dispatch'. The output shows 'Built-in Dispatch Shell' and 'dispatch> help'. Below this, it lists 'Built-in commands:' with a list: 'pwd - show current directory', 'cd - change directory', 'help - show available commands', and 'exit - exit the shell'. Then 'dispatch> pwd' is entered, outputting '/home/monis/OSLAB/skill'. Next, 'dispatch> cd /tmp' is entered. Then 'dispatch> hello' is entered, outputting 'Invalid command: hello'. Then 'dispatch> cd /directory_that_does_not_exist' is entered, outputting 'cd: No such file or directory'. Then 'dispatch> cd' is entered, outputting 'cd: missing argument'. Finally, 'dispatch>' is shown at the prompt.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd  - show current directory
  cd   - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch> cd /tmp
dispatch> hello
Invalid command: hello
dispatch> cd /directory_that_does_not_exist
cd: No such file or directory
dispatch> cd
cd: missing argument
dispatch>
```

Step 16 — Test the exit Command

Now we can finish testing the built-in dispatch program.

At:

dispatch>

type:

exit

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./builtin_dispatch
Built-in Dispatch Shell
dispatch> help
Built-in commands:
  pwd  - show current directory
  cd   - change directory
  help - show available commands
  exit - exit the shell
dispatch> pwd
/home/monis/OSLAB/skill
dispatch> cd /tmp
dispatch> hello
Invalid command: hello
dispatch> cd /directory_that_does_not_exist
cd: No such file or directory
dispatch> cd
cd: missing argument
dispatch> exit
Dispatch shell exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```